

Day 11 Plan

Maths: Mixture & Alligation, Chain Rule · Reasoning: Scheduling & Floor Puzzles + Syllogism/Venn · Science: Chemistry 3: Metals, Non-metals, Periodic Table & Alloys

Today's schedule (about 5½ hours with breaks)

#	Block	Time	What you do	Where
0	Day 10 recall	15 min	10 questions from Day 10, no notes. Redo yesterday's mistakes.	Page 1-2
1	Maths	75 min	5 simplification questions for speed → formulas & methods (10 min) → 2-3 book questions per pattern, timed → 10 Testbook PYQs on this topic → log mistakes	Section A · book · Testbook
-	Break	10 min	Away from the screen	-
2	Reasoning	45 min	Learn the method (10 min) → 15 timed questions, Testbook PYQs first → log	Section B · Testbook
-	Break	10 min	Away from the screen	-
3	Science	110 min	7 lessons with quick checks (≈60 min) → 20-question practice set → Testbook PYQs: metals & non-metals, periodic table (A tier)	Section C · Testbook
-	Break	10 min	Away from the screen	-
4	General awareness	45 min	20 current-affairs MCQs (20 min) → static GA: Economy 2 — banking, NITI Aayog, government schemes — Testbook GA PYQs on it (25 min)	CA PDF · Testbook
5	Close	15 min	Redo today's wrong questions · write 3 rules for tomorrow · send me your scores	Mistake log

Short day? Cut in this order: Testbook science PYQs → reasoning down to 10 questions → static GA down to 10 min. Never cut recall, maths or the science lessons. · Exam pace (average 54 sec a question): GA ~30 sec · Science ~45 · Reasoning ~50 · Maths ~65.

Block 0 — Day 10 recall (answers on the last page)

#	Question	Your answer
1	Type of reaction: $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$?	
2	Chemical formula of rust?	
3	Acid in tamarind?	
4	Colour of phenolphthalein in a base?	
5	pH of pure water?	
6	Formula of baking soda?	
7	Formula of plaster of Paris?	
8	Ratio in aqua regia?	
9	Gas used to flush chip packets?	
10	Maths: $72 \text{ km/h} = ? \text{ m/s}$	

Fewer than 8 right? Re-read only the lessons you missed — not the whole chapter.

A · MATHS — Mixture & Alligation, Chain Rule

Core idea

Alligation is a shortcut for mixing two things at different prices (or strengths) to get a mean value. Write the cheaper and the dearer value on top, the mean in the middle, and cross-subtract: the differences give the **ratio of quantities**. The **chain rule** is proportion: decide for each quantity whether it moves the answer up (direct) or down (inverse).

Formulas

Concept	Formula / rule
Alligation	Cheaper : dearer = (dearer – mean) : (mean – cheaper)
Water is free	Treat water as price 0 in the alligation
Repeated replacement	Pure left = initial $\times (1 - x/C)^n$ (x removed and replaced n times from C)
Mixing two mixtures	Convert each to a fraction of the ingredient, then alligate or average
Chain rule	More men $\rightarrow$ fewer days (inverse) · more work $\rightarrow$ more days (direct)
Work equivalence	$M_1D_1H_1/W_1 = M_2D_2H_2/W_2$

Patterns — the method and one solved example each

Pattern	Method	Solved example
1. Two prices	Alligation	₹40 and ₹60 rice mixed to ₹52 $\rightarrow (60 - 52) : (52 - 40) = 8 : 12 = \mathbf{2 : 3}$
2. Add water	Water at ₹0	30 L milk at ₹20; want ₹16 per litre $\rightarrow$ milk : water = $16 : 4 = 4 : 1 \rightarrow$ water 7.5 L
3. Gain by adding water	Gain % $\rightarrow$ water share	Sells at milk's cost price, gains 25% $\rightarrow$ water : milk = 1 : 4
4. Replacement	$(1 - x/C)^n$	40 L milk; 4 L taken out and replaced by water, twice $\rightarrow 40 \times 0.9^2 = \mathbf{32.4 L}$ milk
5. Two mixtures	Fractions	3 : 1 and 5 : 3 milk-water mixed equally $\rightarrow$ milk $\frac{3}{4}$ and $\frac{5}{8} \rightarrow$ average $11/16 \rightarrow \mathbf{11 : 5}$
6. Mean profit %	Alligate the %	Part sold at 10% gain, rest at 20%; overall 14% $\rightarrow (20 - 14) : (14 - 10) = \mathbf{3 : 2}$
7. Chain rule	$M_1D_1/W_1 = M_2D_2/W_2$	15 men make 30 toys in 6 days $\rightarrow$ 20 men, 40 toys: $15 \times 6 \times 40 \div (30 \times 20) = \mathbf{6 days}$
8. With hours	$M_1D_1H_1 = M_2D_2H_2$	10 men $\times$ 8 h $\times$ 12 days = 16 men $\times$ 6 h $\times$ D $\rightarrow D = \mathbf{10 days}$
9. Food stock	Remaining food $\div$ people	Food for 500 men for 30 days; after 10 days 100 leave $\rightarrow 500 \times 20 \div 400 = \mathbf{25 days}$

Traps

- The mean must lie **between** the two values, or the question is about something else.
- Alligation gives the ratio of **quantities**; multiply back to get actual litres or kilograms.
- Replacement formula only when the **same amount** is removed each time.

B · REASONING — Scheduling & Floor Puzzles + Syllogism/Venn

Part 1 · Scheduling and floor puzzles (S tier — 40 min)

Method: draw a grid (days Mon–Sun, or floors 1 at the bottom upward). Place the **definite** clues first ("R is on Wednesday"), then the relative ones ("immediately after", "before", "between"). Fill the last slot by elimination.

Worked example — weekday puzzle

P, Q, R, S, T each attend a class on a different day, Monday to Friday. R attends on Wednesday. T attends before R but not on Monday. P attends the day immediately after Q.

R = Wed → T = Tue (before Wed, not Mon) → Q, P must be consecutive among Mon, Thu, Fri → Q = Thu, P = Fri → S = Mon.

Mon S · Tue T · Wed R · Thu Q · Fri P

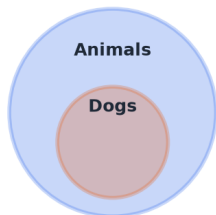
Worked example — floor puzzle

A–E live on floors 1–5 (1 = bottom), one per floor. C is on the top floor. A is immediately below C. B is on an odd floor but not floor 1. D lives below E.

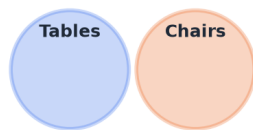
C = 5, A = 4, B = 3 → floors 1 and 2 left; D below E → **D = 1, E = 2**

Part 2 · Syllogism and Venn diagrams (B tier — 20 min only)

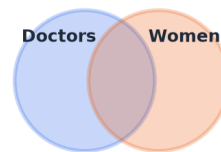
All Dogs are Animals



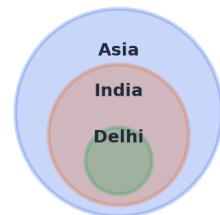
No Table is a Chair



Some Doctors are Women



Delhi ⊂ India ⊂ Asia



Statements	Definite conclusion
All A are B + All B are C	All A are C (and Some C are A)
All A are B + No B is C	No A is C
Some A are B + All B are C	Some A are C
Some A are B + No B is C	Some A are not C
All A are B → converse	Some B are A (NOT all B are A)

Draw the **smallest** diagram that fits the statements. A conclusion follows only if it is true in **every** possible diagram; 'possible but not certain' does not follow.

Patterns, with solved examples

Pattern	Example	Solution
Weekday	In the puzzle above, who attends on Friday?	P
Floor	In the floor puzzle, who lives on floor 2?	E
Syllogism	All pens are books. All books are bags. I: All pens are bags. II: Some bags are pens.	Both follow
Syllogism	Some cats are dogs. No dog is a rat. I: Some cats are not rats.	Follows
Syllogism trap	All roses are flowers. Some flowers are red. I: Some roses are red.	Does not follow (possible, not certain)
Venn	Teachers, Men, Fathers	Fathers inside Men; Teachers overlaps both

Speed rule

In puzzles, the negative clues ('not on Monday') usually decide the answer — mark them with a cross in the grid.

C · SCIENCE — Chemistry 3: Metals, Non-metals, Periodic Table & Alloys

7 lessons. Metals and non-metals were 13 of 210 science questions (R); periodic-table questions are quick recall. Learn the reactivity series as a single string: K Na Ca Mg Al Zn Fe Pb H Cu Hg Ag Au.

Lesson 1 · Physical properties and the famous exceptions (about 8 min)

Property	Metals	Non-metals
Lustre	Shiny	Dull (exception: iodine is lustrous)
Malleable / ductile	Yes — gold is the most malleable and most ductile	Brittle
Conduct heat and electricity	Yes — silver best, then copper; lead is poor	No (exception: graphite conducts)
State at room temperature	Solid (exception: mercury is liquid)	Solid or gas (exception: bromine is liquid)
Hardness	Hard (exception: sodium, potassium are soft — cut with a knife)	Soft (exception: diamond , hardest natural substance)
Melting point	High (exception: gallium, caesium melt in your palm)	Low

Storage: **sodium and potassium in kerosene** (they react violently with air and water); **white phosphorus in water** (catches fire in air).

Quick check 1 — answer out loud before moving on (answers at the end)

a) A metal and a non-metal that are liquid at room temperature? b) Why is sodium kept in kerosene?

Lesson 2 · Chemical properties and the reactivity series (about 10 min)

reactivity decreases ↓	K	react with cold water violently	A metal displaces any metal BELOW it from its salt solution. $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu} \quad \checkmark$ $\text{Cu} + \text{FeSO}_4 \rightarrow \text{no reaction}$
	Na		
	Ca	react with cold water	
	Mg	hot water	
	Al	steam (Al, Zn, Fe)	
	Zn		Extraction: K → Al: electrolysis of molten ore Zn → Pb: reduce oxide with carbon Cu, Hg: heat alone / easy Au, Ag: found free (native)
	Fe		
	Pb		
	H	hydrogen — reference	
	Cu	no H ₂ with dilute acids	
	Hg		
	Ag		
	Au	least reactive	

Reaction	Result
Metal + oxygen	Basic oxide (Na ₂ O, MgO). Amphoteric oxides react with acids and bases: Al ₂ O ₃ , ZnO
Non-metal + oxygen	Acidic oxide (CO ₂ , SO ₂) or neutral oxide (CO, N ₂ O)
Metal + water	K, Na violently with cold water · Ca less · Mg with hot water · Al, Zn, Fe with steam · Pb, Cu, Ag, Au not at all
Metal + dilute acid	Salt + hydrogen — only metals above H. With HNO ₃ , H ₂ is usually not given off (it oxidises H ₂), except Mg and Mn with very dilute HNO ₃

Quick check 2 — answer out loud before moving on (answers at the end)

a) Will copper displace iron from FeSO₄ solution? b) Name an amphoteric oxide.

Lesson 3 · Ionic and covalent bonds (about 8 min)

Ionic bond: a metal gives electrons to a non-metal. Na (2, 8, 1) loses 1 → Na⁺; Cl (2, 8, 7) gains 1 → Cl⁻; the opposite charges attract → NaCl. Mg gives 2 electrons to two Cl atoms → MgCl₂.

Covalent bond: non-metal atoms **share** electron pairs. H₂ single bond · O₂ double bond · N₂ **triple** bond · CH₄, H₂O, NH₃.

	Ionic compounds	Covalent compounds
State	Hard, brittle solids	Often liquids or gases
Melting / boiling point	High	Low
Solubility	In water, not in kerosene or petrol	Usually in organic solvents
Conduct electricity	When molten or dissolved (ions move), not as solids	Generally no

Quick check 3 — answer out loud before moving on (answers at the end)

a) Why does solid NaCl not conduct electricity, while its solution does? b) Type of bond in N₂?

Lesson 4 · From ore to metal (metallurgy) (about 9 min)

A **mineral** is any natural compound of a metal; an **ore** is a mineral from which the metal can be extracted profitably. The unwanted earthy material is **gangue**.

Step	What happens
Concentration	Remove gangue — froth flotation for sulphide ores
Roasting	Sulphide ore heated in excess air → oxide ($2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$)
Calcination	Carbonate ore heated in limited air → oxide ($\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$)
Reduction	Oxide → metal: carbon for Zn, Fe, Pb · electrolysis of the molten compound for Na, Mg, Al · heat alone for Hg (from cinnabar)
Refining	Electrolytic refining: impure metal = anode, pure metal = cathode; impurities fall as anode mud

Thermite reaction: $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow 2\text{Fe} + \text{Al}_2\text{O}_3$ + lots of heat — molten iron used to **weld railway tracks** and cracked machine parts.

Metal	Ore(s)
Aluminium	Bauxite (cryolite used in extraction)
Iron	Haematite, magnetite
Copper	Copper pyrites (chalcopyrite)
Zinc	Zinc blende, calamine
Lead	Galena
Mercury	Cinnabar

Quick check 4 — answer out loud before moving on (answers at the end)

a) What is the thermite reaction used for on the railways? b) Main ore of aluminium?

Lesson 5 · Alloys (about 7 min)

An **alloy** is a homogeneous mixture of a metal with other metals or a non-metal. Alloys are harder and resist corrosion better, but have lower electrical conductivity and melting point than the pure metal (so **solder**, with its low melting point, joins wires).

Alloy	Made of	Alloy	Made of
Brass	Cu + Zn	Steel	Fe + C
Bronze	Cu + Sn	Stainless steel	Fe + Cr + Ni (+ C)
Solder	Pb + Sn	Duralumin	Al + Cu + Mg + Mn (aircraft)
German silver	Cu + Zn + Ni (no silver!)	Magnalium	Al + Mg
Gun metal	Cu + Sn + Zn	Amalgam	Any metal + mercury

Pure gold = 24 carat. 22-carat gold = 22 parts gold + 2 parts copper or silver (pure gold is too soft for jewellery).

Quick check 5 — answer out loud before moving on (answers at the end)

a) Composition of bronze? b) Does German silver contain silver?

Lesson 6 • The periodic table (about 10 min)

Scientist	Idea	Remember
Döbereiner	Triads: the middle element's mass $\approx$ average of the other two	Li, Na, K · Ca, Sr, Ba · Cl, Br, I
Newlands	Law of octaves: every 8th element is similar (like musical notes)	Worked only up to calcium
Mendeleev	Arranged 63 elements by atomic mass ; left gaps for undiscovered ones	Eka-boron = scandium, eka-aluminium = gallium, eka-silicon = germanium
Moseley (modern law)	Properties depend on atomic number	Modern table: 18 groups, 7 periods, 118 elements
Trend	Across a period (left → right)	Down a group
Atomic size	Decreases	Increases
Metallic character	Decreases	Increases
Electronegativity	Increases	Decreases
Valency	1 → 4 → 1 (by outer electrons)	Same in the group

Group names: **1** alkali metals · **2** alkaline earth metals · **17** halogens · **18** noble gases (inert). Lanthanides and actinides form the separate f-block rows at the bottom.

Quick check 6 — answer out loud before moving on (answers at the end)

a) Which element was eka-aluminium? b) Down a group, does atomic size increase or decrease?

Lesson 7 • Must-know element facts (about 6 min)

Fact	Answer
Most abundant element in the universe	Hydrogen
Most abundant in the Earth's crust	Oxygen (then silicon)
Most abundant metal in the crust	Aluminium
Most abundant gas in air	Nitrogen $\approx$ 78% (oxygen $\approx$ 21%)
Lightest element / gas	Hydrogen
Gas filled in filament bulbs	Argon (inert)
Gas in advertising sign lights (red glow)	Neon
Gas in weather balloons; with O ₂ for divers	Helium
Allotropes of carbon	Diamond (hardest), graphite (lubricant, pencil 'lead', electrodes), fullerene C ₆₀
Allotrope of oxygen	Ozone, O ₃

Quick check 7 — answer out loud before moving on (answers at the end)

a) Most abundant metal in the Earth's crust? b) Gas filled in filament bulbs?

Science 1-page summary — revise from this tomorrow

Topic	Must remember
Exceptions	Liquid: Hg (metal), Br (non-metal) · soft: Na, K · melt in palm: Ga, Cs · lustrous non-metal: I · conducting non-metal: graphite
Best	Silver best conductor · gold most malleable and ductile · diamond hardest · Na, K in kerosene, white P in water
Reactivity	K Na Ca Mg Al Zn Fe Pb H Cu Hg Ag Au · higher displaces lower · metals below H give no H ₂ with acids
Oxides	Metal → basic · non-metal → acidic · Al ₂ O ₃ , ZnO amphoteric · CO, N ₂ O neutral
Bonds	Ionic: transfer, high m.p., conduct when molten/dissolved · covalent: sharing; N ₂ triple bond
Metallurgy	Froth flotation · roasting (sulphide, excess air) · calcination (carbonate, limited air) · thermite welds rails
Ores	Bauxite Al · haematite Fe · cinnabar Hg · galena Pb · zinc blende/calamine Zn · copper pyrites Cu
Alloys	Brass Cu+Zn · bronze Cu+Sn · solder Pb+Sn · stainless Fe+Cr+Ni · duralumin aircraft · 24 carat = pure gold
Periodic table	Döbereiner triads · Newlands octaves · Mendeleev mass, gaps (Ga, Ge, Sc) · Moseley atomic number · 18 groups, 7 periods

Science practice set — 20 questions · target 6 minutes

#	Question
1	Which metal is liquid at room temperature? (a) Gallium (b) Sodium (c) Mercury (d) Bromine
2	The best conductor of electricity is: (a) copper (b) gold (c) silver (d) aluminium
3	Which non-metal conducts electricity? (a) Graphite (b) Diamond (c) Sulphur (d) Iodine
4	Sodium is stored in: (a) water (b) alcohol (c) open air (d) kerosene
5	Which metal reacts violently with cold water? (a) Iron (b) Potassium (c) Copper (d) Zinc
6	Which metal does NOT give hydrogen with dilute HCl? (a) Zinc (b) Magnesium (c) Iron (d) Copper
7	Aluminium oxide is: (a) basic only (b) acidic only (c) neutral (d) amphoteric
8	Ionic compounds conduct electricity: (a) as solids (b) never (c) only in kerosene (d) when molten or in solution
9	The bond in a nitrogen molecule is: (a) single covalent (b) double covalent (c) triple covalent (d) ionic
10	Froth flotation is used to concentrate: (a) carbonate ores (b) sulphide ores (c) oxide ores only (d) native metals
11	Heating a carbonate ore in limited air is called: (a) calcination (b) roasting (c) smelting (d) refining
12	Railway tracks are welded using the: (a) Haber process (b) contact process (c) electrolysis of brine (d) thermite reaction
13	Cinnabar is an ore of: (a) lead (b) zinc (c) mercury (d) iron
14	Bronze is an alloy of: (a) copper and zinc (b) lead and tin (c) iron and carbon (d) copper and tin
15	Solder is an alloy of: (a) lead and tin (b) copper and tin (c) iron and nickel (d) aluminium and magnesium
16	Pure gold is: (a) 22 carat (b) 18 carat (c) 24 carat (d) 20 carat
17	Mendeleev's eka-silicon is now called: (a) gallium (b) germanium (c) scandium (d) silicon
18	The modern periodic table is based on: (a) atomic number (b) atomic mass (c) number of neutrons (d) density
19	Group 17 elements are called: (a) noble gases (b) alkali metals (c) alkaline earth metals (d) halogens
20	The most abundant metal in the Earth's crust is: (a) iron (b) calcium (c) aluminium (d) sodium

Score	Time taken	Wrong Q numbers	Mistake type (concept / formula / calc / careless / recall)
/ 20			

Answers

Science practice set

#	Ans	Why / lesson to re-read
1	c	Lesson 1 (bromine is a non-metal)
2	c	Lesson 1
3	a	Lesson 1
4	d	Lesson 1
5	b	Lesson 2
6	d	Lesson 2 — below H
7	d	Lesson 2
8	d	Lesson 3
9	c	Lesson 3
10	b	Lesson 4
11	a	Lesson 4
12	d	Lesson 4
13	c	Lesson 4
14	d	Lesson 5
15	a	Lesson 5
16	c	Lesson 5
17	b	Lesson 6
18	a	Lesson 6 — Moseley
19	d	Lesson 6
20	c	Lesson 7

Quick checks

QC	Answers
1	a) Mercury (metal) and bromine (non-metal) b) It reacts violently with oxygen and moisture in air; kerosene keeps air and water away
2	a) No — copper is below iron in the reactivity series b) Aluminium oxide or zinc oxide
3	a) In the solid the ions are fixed; in solution they can move and carry charge b) Triple covalent bond
4	a) Welding rail tracks (molten iron fills the joint) b) Bauxite
5	a) Copper + tin b) No — it is copper, zinc and nickel
6	a) Gallium b) Increases
7	a) Aluminium b) Argon

Day 10 recall (Block 0)

#	Answer	#	Answer
1	Combination	6	NaHCO_3
2	$\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$	7	$\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$
3	Tartaric acid	8	3 HCl : 1 HNO_3
4	Pink	9	Nitrogen
5	7	10	20 m/s

Send me tonight

Day 10 recall score /10 · science practice score /20 and time · Maths and Reasoning: attempted, correct, average seconds per question · the question numbers you got wrong.